

Q1 2021 (01 Sep Shift 2)

Consider the parabola with vertex $\left(\frac{1}{2}, \frac{3}{4}\right)$ and the directrix $y = \frac{1}{2}$. Let P be the point where the parabola meets the line $x = -\frac{1}{2}$. If the normal to

the parabola at P intersects the parabola again at

the point Q , then $(PQ)^2$ is equal to :

(1) $\frac{75}{8}$

(2) $\frac{125}{16}$

(3) $\frac{25}{2}$

(4) $\frac{15}{2}$

Q2 2021 (31 Aug Shift 2)

A tangent line L is drawn at the point $(2, -4)$ on the parabola $y^2 = 8x$. If the line L is also tangent to the circle $x^2 + y^2 = a$, then ' a ' is equal to ,

Q3 2021 (31 Aug Shift 1)

The length of the latus rectum of a parabola, whose vertex and focus are on the positive x -axis at a distance R and $S(> R)$ respectively from the origin, is:

(1) $4(S + R)$

(2) $2(S - R)$

(3) $4(S - R)$

(4) $2(S + R)$

Q4 2021 (27 Aug Shift 2)

If two tangents drawn from a point P to the parabola $y^2 = 16(x - 3)$ are at right angles, then the locus of point P is :

(1) $x + 3 = 0$

$$(2) x + 1 = 0$$

$$(3) x + 2 = 0$$

$$(4) x + 4 = 0$$

Q5 2021 (27 Aug Shift 1)

A tangent and a normal are drawn at the point $P(2, -4)$ on the parabola $y^2 = 8x$, which meet the directrix of the parabola at the points A and B respectively. If $Q(a, b)$ is a point such that $AQBP$ is a square, then $2a + b$ is equal to :

$$(1) -16$$

$$(2) -18$$

$$(3) -12$$

$$(4) -20$$

Answer Key

Q1 (2)

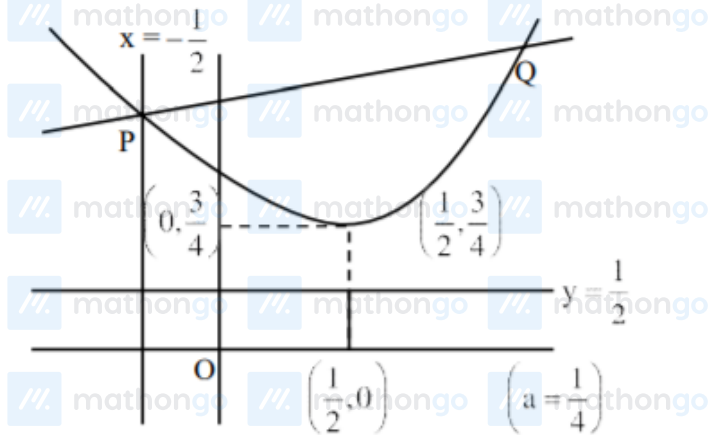
Q2 (2)

Q3 (3)

Q4 (2)

Q5 (1)

Q1 (2)



$$\left(y - \frac{3}{4}\right) = \left(x - \frac{1}{2}\right)^2 \dots (1)$$

For $x = -\frac{1}{2}$

$$y - \frac{3}{4} = 1 \Rightarrow y = \frac{7}{4} \Rightarrow P\left(-\frac{1}{2}, \frac{7}{4}\right)$$

$$\text{Now } y' = 2\left(x - \frac{1}{2}\right) \quad \text{At } x = -\frac{1}{2}$$

$$\Rightarrow m_T = -2, m_N = \frac{1}{2}$$

Equation of Normal is

$$y - \frac{7}{4} = \frac{1}{2}\left(x + \frac{1}{2}\right)$$

$$y = \frac{x}{2} + 2$$

Now put y in equation (1)

$$\frac{x}{2} + 2 - \frac{3}{4} = \left(x - \frac{1}{2}\right)^2$$

$$\Rightarrow x = 2 \text{ or } -\frac{1}{2}$$

$$\Rightarrow Q(2, 3)$$

$$\text{Now } (PQ)^2 = \frac{125}{16}$$

Option (2)

Q2 (2)

tangent of $y^2 = 8x$ is $y = mx + \frac{2}{m}$

$$P(2, -4) \Rightarrow -4 = 2m + \frac{2}{m}$$

$$\Rightarrow m + \frac{1}{m} = -2 \Rightarrow m = -1$$

$\therefore$ tangent is $y = -x - 2$

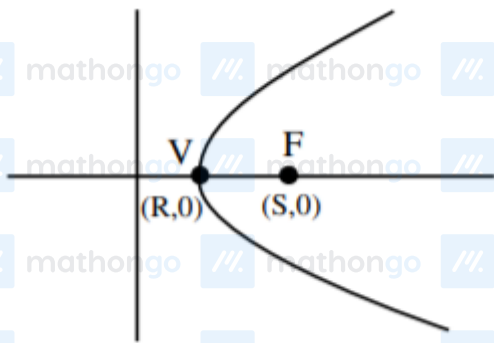
$$\Rightarrow x + y + 2 = 0 \dots (1)$$

(1) is also tangent to $x^2 + y^2 = a$

$$\text{So } \frac{2}{\sqrt{2}} = \sqrt{a} \Rightarrow \sqrt{a} = \sqrt{2}$$

$$\Rightarrow a = 2$$

Q3 (3)



$V \rightarrow$ Vertex

$F \rightarrow$ focus

$$VF = S - R$$

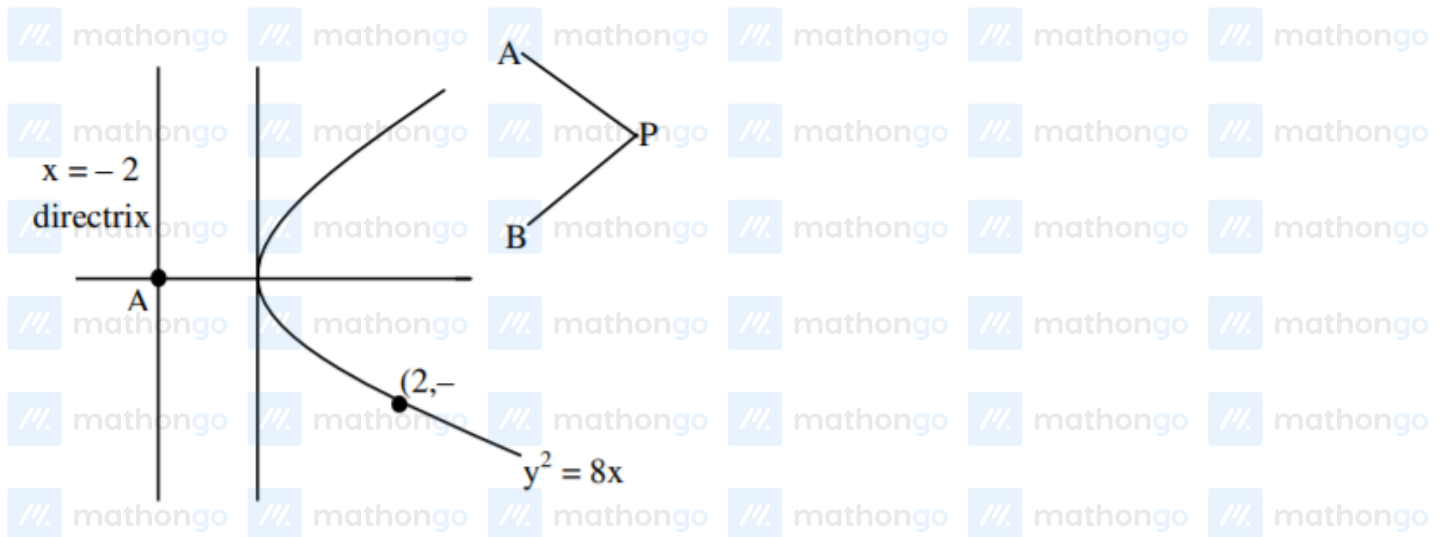
$$\text{So latus rectum} = 4(S - R)$$

Q4 (2)

Locus is directrix of parabola

$$x - 3 + 4 = 0 \Rightarrow x + 1 = 0$$

Q5 (1)



Equation of tangent at $(2, -4)$ ($T = 0$)

$$-4y = 4(x + 2)$$

$$x + y + 2 = 0 \dots (1)$$

equation of normal

$$x - y + \lambda = 0$$

$$\downarrow (2, -4)$$

$$\lambda = -6$$

thus $x - y = 6 \dots (2)$ equation of normal

POI of (1) & $x = -2$ is $A(-2, 0)$

POI of (2) & $x = -2$ is $A(-2, 8)$

Given AQB P is a sq.



$$\Rightarrow m_{AQ} \cdot m_{AP} = -1$$

$$\Rightarrow \left(\frac{b}{a+2}\right) \left(\frac{4}{-4}\right) = -1 \Rightarrow a + 2 = b \dots (1)$$

Also PQ must be parallel to x -axis thus

$$\Rightarrow b = -4$$

$$\therefore a = -6$$

$$\text{Thus } 2a + b = -16$$